

Digital Image Processing Assignment 01 – Report Analysis

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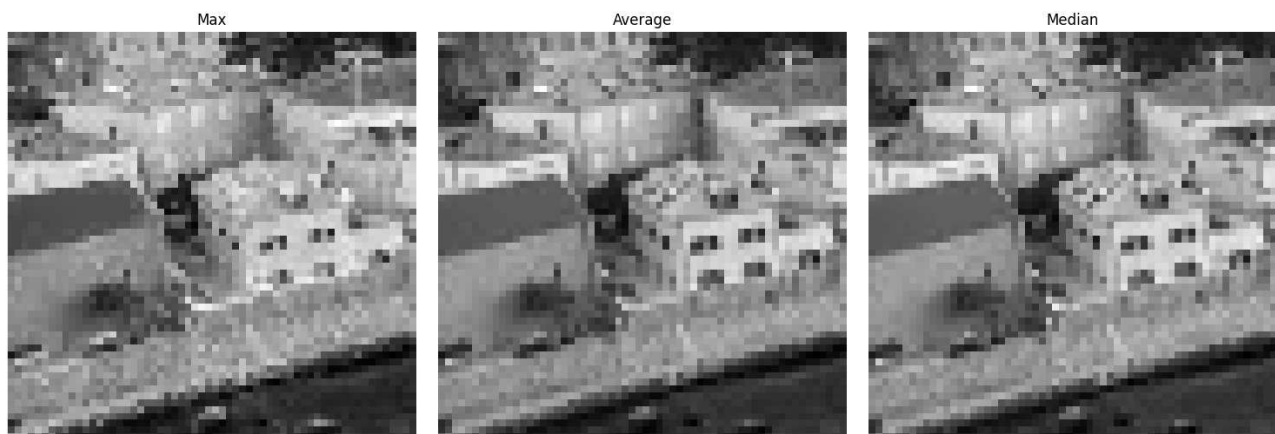
1. Downsampling

In this experiment, three different downsampling methods were tested, which are Max Pooling, Average Pooling, and Median Pooling. The methods were tested using images with different characteristics, including natural scenes, text and edges, and fine details. Both grayscale and color images were also used to see how each method affects the image.

In the image that has gone through Max Pooling, the bright areas become more prominent compared to the other methods. The image also looks more 'crisp' in some areas, but the pixelated effect is more apparent. This happens because Max Pooling keeps the highest pixel value from each surrounding pixel area. As a result, bright areas and strong highlights are preserved more prominently.

While in Average Pooling and Median Pooling, there are barely any differences at first glance. Both offer a much smoother image than Max Pooling as they provide a more 'in the middle' representation. Of course, Average and Median are two different things, but both represent the surrounding pixel values in a way that reduces the more extreme differences between pixels.

In Average Pooling, the output of each pixel is a calculated average of the neighboring pixels. Hence, it results in a smoother transition. However, fine details and sharp edges can lose their characteristics because high-frequency information is averaged with the surrounding pixels.



In Median Pooling, on the other hand, selects a middle value from the neighboring pixels, so it is less affected by extreme pixel values when an image contains noise or isolated bright or dark pixels. However, similar to Average Pooling, it can also lose some fine details in the process.

The same methods were also tested on color images. The overall behavior is similar to the grayscale results, although the differences can be seen through the colors as well. Max Pooling can make some colors appear brighter or more intense, while Average Pooling produces smoother color

transitions. Median Pooling provides a more balanced result and is less affected by extreme color values.

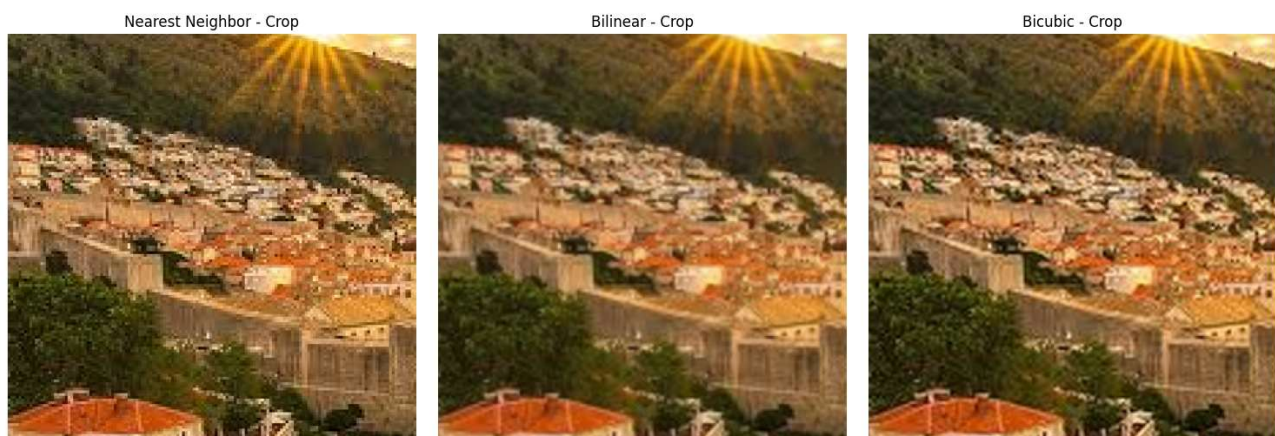
2. Upsampling

After testing the downsampling methods, the Average Pooling result was used as the low-resolution input for the upsampling experiment. Three different methods were tested, which are Nearest Neighbor, Bilinear Interpolation, and Bicubic Interpolation.

In the Nearest Neighbor result, the pixels are much more apparent and the image looks more blocky compared to the other methods. This happens because each new pixel takes the value of the nearest existing pixel. Because there is no calculation between surrounding pixels, the original pixel blocks become more visible when the image is enlarged.

In Bilinear Interpolation, the image looks smoother than Nearest Neighbor. It calculates the new pixel values based on the surrounding pixels, creating a smoother transition between them. However, this can also make some edges and fine details appear slightly blurred.

Bicubic Interpolation produces a smoother result compared to Nearest Neighbor and Bilinear. It uses a larger group of surrounding pixels to calculate the new pixel values, allowing the image to have smoother transitions and more refined edges. However, even though the result looks smoother, it cannot restore details that were already lost during downsampling.



Conclusion

From this experiment, each method produces a different result depending on how the pixel values are selected or calculated. Max Pooling makes bright areas more prominent, Average Pooling produces a smoother image, and Median Pooling is less affected by extreme pixel values.

For upsampling, Nearest Neighbor produces the most noticeable pixelation, while Bilinear produces a smoother result. Bicubic gives a smoother and more refined appearance among the three. However, upsampling can only estimate the missing pixels, so it cannot fully restore information that was already lost during downsampling.